THE COMPARISON MAP OF A PROPER MORPHISM IS AN ISOMORPHISM OVER THE OPEN LOCUS WHERE IT IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6Q (sources/stacks-duality/pinned-original-source.tex line 2171).

Allowed standard prerequisites (editorial).

- Sheafiness of the right adjoint in the Noetherian proper case. Same chapter, Lemma labelled lemma-proper-noetherian, cited at proof lines 2180-2182. Gives that (equation-sheafy) is an isomorphism.
- Restriction of the comparison map. Same chapter, Lemma labelled lemma-restriction-compare-with-pullback, cited at proof lines 2182-2185. Identifies the restricted map with the comparison map of the restricted situation.

Source reference. Original equation 0A9S: sources/stacks-duality/pinned-original-source.tex, source lines 2079–2079. Same-chapter pointer into the author environment equation at source line 2079; invoked at proof line 2182 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9P: sources/stacks-duality/pinned-original-source.tex, source lines 1106–1106. Same-chapter pointer into the author environment lemma at source line 1106; invoked at proof line 2179 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0A9L: sources/stacks-duality/pinned-original-source.tex, source lines 988–988. Same-chapter pointer into the author environment equation at source line 988; invoked at proof line 2179 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0B6P: sources/stacks-duality/pinned-original-source.tex, source lines 2119–2119. Same-chapter pointer into the author environment lemma at source line 2119; invoked at proof line 2181 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of Noetherian schemes. Let $V \subset Y$ be an open such that $f^{-1}(V) \to V$ is an isomorphism. Then for $K \in D^+_{QCoh}(\mathcal{O}_Y)$ the map (0A9S) restricts to an isomorphism over $f^{-1}(V)$.*

Proof. By Lemma 0A9P the map (0A9L) is an isomorphism for objects of $D^+_{QCoh}(\mathcal{O}_Y)$. Hence Lemma 0B6P tells us the restriction of (0A9S) for K to $f^{-1}(V)$ is the map (0A9S) for $K|_V$ and $f^{-1}(V) \to V$. Thus it suffices to show that the map is an isomorphism when f is the identity morphism. This is clear. □